CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD

CENTRAL VALLEY REGION
11020 SUN CENTER DR #200
RANCHO CORDOVA CA 95670-6114

2023 ANNUAL REPORT

PREPARED FOR:

SOARES DAIRY FARMS INC. – MERCEY SPRINGS DAIRY

21750 S MERCEY SPRINGS RD

LOS BANOS CA 93635

NOTES:

PREPARED BY:

CARDOSO AG SERVICES

PO BOX 906
NEWMAN CA 95360

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Soares Dairy Farms Inc. - Mercy Springs

Physical address of dairy:

Number and Street	City	County	Zip Code
21750 S Mercey Springs RD	Los Banos	Merced	93635

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 04/01/1940

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X088-X103-X009-XXXX

B. OPERATORS

Soares, David

Operator name: Soares, David

Telephone no.: (209) 826-3626 Landline (209) 509-8436 Cellular

Mailing Address Number and Street	City	State	Zip Code
14515 Badger Flat Road	Los Banos	CA	93635

Soares Dairy Farms Inc.

Operator name: Soares Dairy Farms Inc.

Telephone no.: (209) 862-3626 Landline Cellular

Mailing Address Number and Street	City	State	Zip Code
14515 S Badger Flat Rd	Los Banos	CA	93635

This operator is responsible for paying permit fees.

C. OWNERS

Soares, David

Legal owner name: Soares, David

Telephone no.: (209) 826-3626 Landline (209) 509-8436 Cellular

Mailing Address Number and Street	City	State	Zip Code
14515 Badger Flat Road	Los Banos	CA	93635

Soares Dairy Farms Inc.

Legal owner name:	Soares Dairy Farms Inc.	Telephone no.:	(209) 862-3626	
			Landline	Cellular
14515 S Badger Flat Rd	Los Banos		CA	93635
Mailing Address Number and Street	City		State	Zip Code

This owner is responsible for paying permit fees.

Soares Living Trust

Legal owner name:	Soares Living Trust	Telephone no.:	(209) 862-3626	
			Landline	Cellular
14515 S Badger Flat Rd	Los Banos		CA	93635
Mailing Address Number and Street	City		State	Zip Code

Annual Report - General Order No. R5-2007-0035
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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	378	52	48	32	35	40
Number under roof	0	0	0	0	0	0
Maximum number	378	52	48	32	35	40
Average number	378	52	48	32	35	40
Avg live weight (lbs)	1,400	1,500	1,100	775		

Predominant milk cow breed: Holstein

Average milk production: 71 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 11,599.55 *tons per reporting period*

Total nitrogen from manure: 148,268.20 *lbs per reporting period*

After ammonia losses (30% loss applied): 103,787.74 *lbs per reporting period*

Total phosphorus from manure: 24,377.16 *lbs per reporting period*

Total potassium from manure: 72,638.83 *lbs per reporting period*

Total salt from manure: 189,938.70 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 4,320,000 *gallons*

Total nitrogen generated: 5,021.82 *lbs*

Total phosphorus generated: 1,631.28 *lbs*

Total potassium generated: 11,536.13 *lbs*

Total salt generated: 74,173.70 *lbs*

	4,320,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	4,320,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
CCID Canal	Surface water
Domestic Well	Ground water
Irrigation Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
MS1	25	25	1	manure	X088-X103-X009-XXXX X088-X103-X010-XXXX
MS2	54	54	2	both	X088-X103-X009-XXXX X088-X103-X010-XXXX
MS-Almond	114	109	2	manure	X088-X084-X006-XXXX X088-X130-X043-XXXX X088-X130-X044-XXXX
Totals for areas that were used for application	193	188	5		
Totals for areas that were not used for application					
Land application area totals	193	188	5		

B. CROPS AND HARVESTS

MS1

Field name: MS1

10/08/2021: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 25 Plant date: 10/08/2021

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
07/03/2023	190.00 *ton*	Dry-weight		28.3	38,100.00	3,000.00	32,500.00		10.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.60	415.23	32.70	354.20	1,155.23

MS2

Field name: MS2

MS2

10/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 54 Plant date: 10/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/22/2023	652.00 *ton*	Dry-weight		51.5	10,400.00	2,300.00	19,300.00		9.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.07	121.80	26.94	226.04	1,124.34

06/10/2023: Corn, silage

Crop: Corn, silage Acres planted: 54 Plant date: 06/10/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/22/2023	1,234.00 *ton*	Dry-weight		69.9	13,100.00	2,400.00	12,600.00		6.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	22.85	180.21	33.02	173.34	839.17

MS-Almond

Field name: MS-Almond

10/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 109 Plant date: 10/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/22/2023	1,362.00 *ton*	Dry-weight		57.9	9,000.00	2,300.00	12,000.00		8.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.50	94.69	24.20	126.25	862.73

MS-Almond

06/10/2023: Corn, silage

Crop: Corn, silage Acres planted: 109 Plant date: 06/10/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/06/2023	2,057.00 *ton*	Dry-weight		56.4	10,400.00	2,600.00	11,900.00		9.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	18.87	171.14	42.79	195.83	1,612.69

NUTRIENT BUDGET

A. LAND APPLICATIONS

MS1 - 10/08/2021: Alfalfa, hay

Field name: MS1

Crop: Alfalfa, hay

Plant date: 10/08/2021

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	2.32	0.00	0.00	662.14	9,918,223.00 *gal*
Application event totals		2.32	0.00	0.00	662.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	2.63	0.00	0.00	752.07	11,265,277.00 *gal*
Application event totals		2.63	0.00	0.00	752.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.95	0.00	0.00	557.71	8,353,903.00 *gal*
Application event totals		1.95	0.00	0.00	557.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.77	0.00	0.00	506.06	7,580,269.00 *gal*
Application event totals		1.77	0.00	0.00	506.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Scraped material	472.73	102.16	481.38	0.00	650.00 *ton*
Application event totals		472.73	102.16	481.38	0.00	

MS2 - 10/08/2022: Oats, silage-soft dough

Field name: MS2

Crop: Oats, silage-soft dough

Plant date: 10/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/05/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	122.97	40.27	200.24	0.00	493.00 *ton*
Application event totals		122.97	40.27	200.24	0.00	

11/16/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	2.57	0.00	0.00	733.80	23,741,815.00 *gal*
Application event totals		2.57	0.00	0.00	733.80	

03/15/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	37.95	14.22	107.15	684.29	2,160,000.00 *gal*
Application event totals		37.95	14.22	107.15	684.29	

MS2 - 06/10/2023: Corn, silage

Field name: MS2

Crop: Corn, silage

Plant date: 06/10/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/23/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Scraped material	181.14	39.15	184.46	0.00	538.00 *ton*
Application event totals		181.14	39.15	184.46	0.00	

MS2 - 06/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	2.14	0.00	0.00	612.42	19,814,720.00 *gal*
Application event totals		2.14	0.00	0.00	612.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	30.13	9.08	55.74	323.79	1,080,000.00 *gal*
CCID Canal	Surface water	1.73	0.00	0.00	493.56	15,969,100.00 *gal*
Application event totals		31.85	9.08	55.74	817.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.52	0.00	0.00	433.80	14,035,427.00 *gal*
Application event totals		1.52	0.00	0.00	433.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	24.92	6.91	50.74	365.51	1,080,000.00 *gal*
CCID Canal	Surface water	1.32	0.00	0.00	378.40	12,242,977.00 *gal*
Application event totals		26.24	6.91	50.74	743.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.25	0.00	0.00	357.58	11,569,450.00 *gal*
Application event totals		1.25	0.00	0.00	357.58	

MS-Almond - 10/08/2022: Oats, silage-soft dough

Field name: MS-Almond

Crop: Oats, silage-soft dough

Plant date: 10/08/2022

MS-Almond - 10/08/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/05/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	122.96	40.26	200.21	0.00	995.00 *ton*
Application event totals		122.96	40.26	200.21	0.00	

11/16/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	2.54	0.00	0.00	727.07	47,483,630.00 *gal*
Application event totals		2.54	0.00	0.00	727.07	

MS-Almond - 06/10/2023: Corn, silage

Field name: MS-Almond

Crop: Corn, silage

Plant date: 06/10/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/23/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Scraped material	224.02	48.41	228.12	0.00	1,343.00 *ton*
Application event totals		224.02	48.41	228.12	0.00	

06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	2.12	0.00	0.00	606.80	39,629,440.00 *gal*
Application event totals		2.12	0.00	0.00	606.80	

06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.71	0.00	0.00	489.04	31,938,200.00 *gal*
Application event totals		1.71	0.00	0.00	489.04	

MS-Almond - 06/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.13	0.00	0.00	322.36	21,053,140.00 *gal*
Application event totals		1.13	0.00	0.00	322.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.97	0.00	0.00	562.39	36,728,930.00 *gal*
Application event totals		1.97	0.00	0.00	562.39	

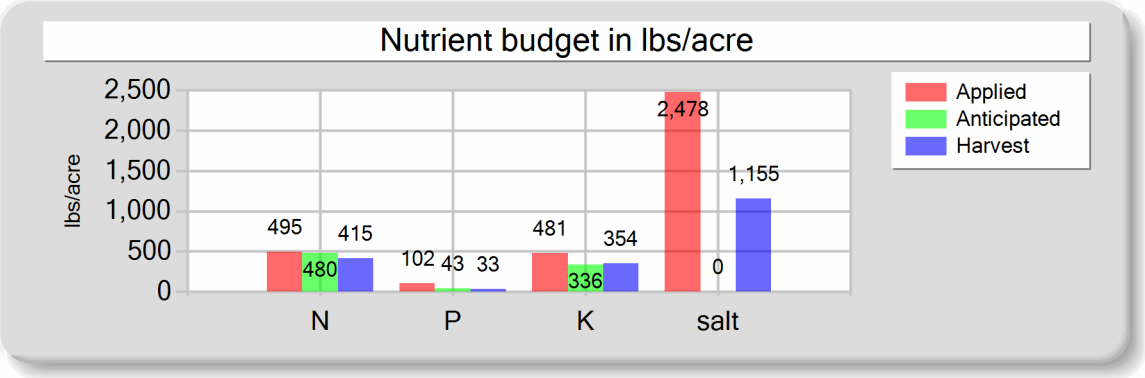
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.24	0.00	0.00	354.30	23,138,900.00 *gal*
Application event totals		1.24	0.00	0.00	354.30	

B. NUTRIENT BUDGET

MS1 - 10/08/2021: Alfalfa, hay

Field name: MS1 Crop: Alfalfa, hay Plant date: 10/08/2021



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	472.73	102.16	481.38	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.67	0.00	0.00	2,477.98
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	495.40	102.16	481.38	2,477.98
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	415.23	32.70	354.20	1,155.23
Nutrient balance	80.17	69.47	127.19	1,322.75
Applied to removed ratio	1.19	3.12	1.36	2.15

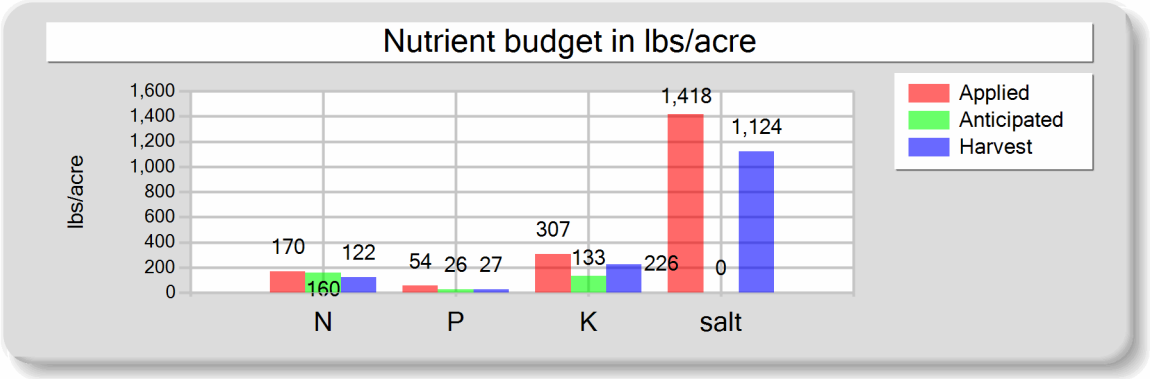
Fresh water applied	
37,117,672.00	*gallons*
1,366.92	*acre-inches*
54.68	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

MS2 - 10/08/2022: Oats, silage-soft dough

Field name: MS2 Crop: Oats, silage-soft dough Plant date: 10/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	122.97	40.27	200.24	0.00
Process wastewater	37.95	14.22	107.15	684.29
Fresh water	2.57	0.00	0.00	733.80
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	170.49	54.49	307.39	1,418.09
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	121.80	26.94	226.04	1,124.34
Nutrient balance	48.69	27.55	81.35	293.75
Applied to removed ratio	1.40	2.02	1.36	1.26

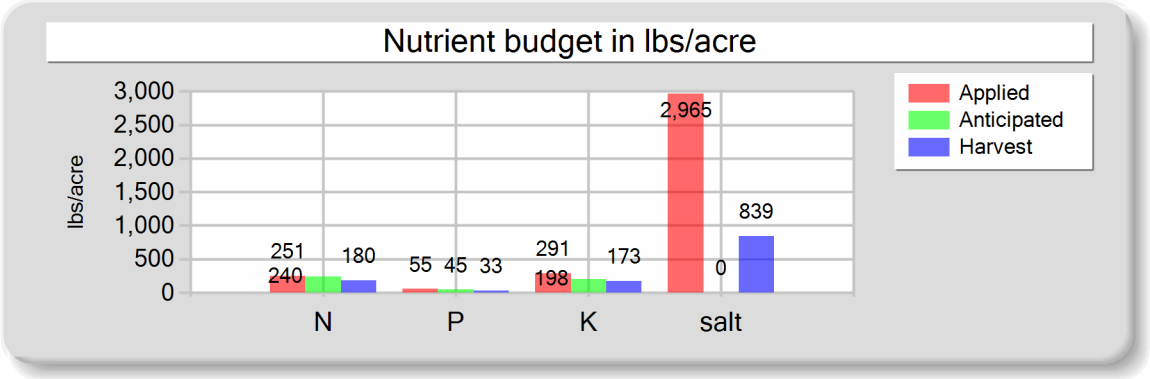
Fresh water applied	
23,741,815.00	*gallons*
874.33	*acre-inches*
16.19	*inches/acre*

Process wastewater applied	
2,160,000.00	*gallons*
79.55	*acre-inches*
1.47	*inches/acre*

Total harvests for the crop	
1	*harvests*

MS2 - 06/10/2023: Corn, silage

Field name: MS2 Crop: Corn, silage Plant date: 06/10/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	181.14	39.15	184.46	0.00
Process wastewater	55.04	15.99	106.48	689.30
Fresh water	7.97	0.00	0.00	2,275.76
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	251.15	55.14	290.94	2,965.06
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	180.21	33.02	173.34	839.17
Nutrient balance	70.94	22.12	117.61	2,125.90
Applied to removed ratio	1.39	1.67	1.68	3.53

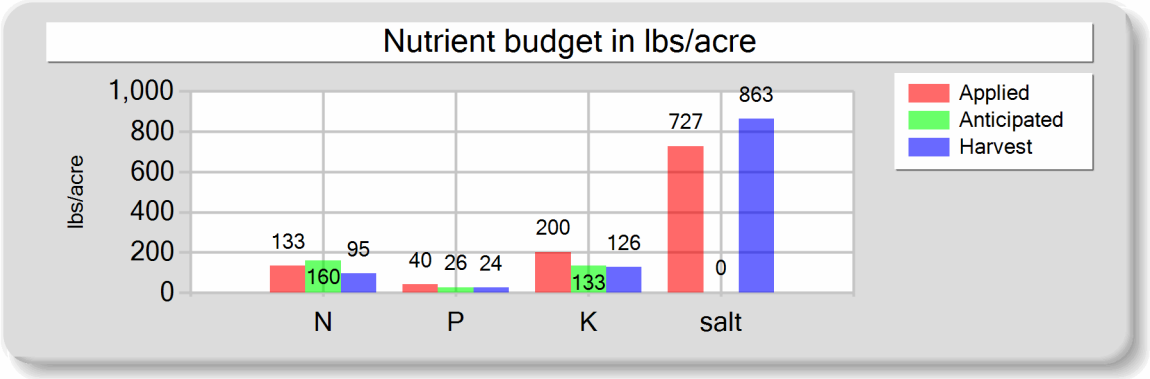
Fresh water applied	
73,631,674.00	*gallons*
2,711.60	*acre-inches*
50.21	*inches/acre*

Process wastewater applied	
2,160,000.00	*gallons*
79.55	*acre-inches*
1.47	*inches/acre*

Total harvests for the crop	
1	*harvests*

MS-Almond - 10/08/2022: Oats, silage-soft dough

Field name: MS-Almond | Crop: Oats, silage-soft dough | Plant date: 10/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	122.96	40.26	200.21	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	2.54	0.00	0.00	727.07
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	132.50	40.26	200.21	727.07
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	94.69	24.20	126.25	862.73
Nutrient balance	37.81	16.06	73.96	-135.67
Applied to removed ratio	1.40	1.66	1.59	0.84

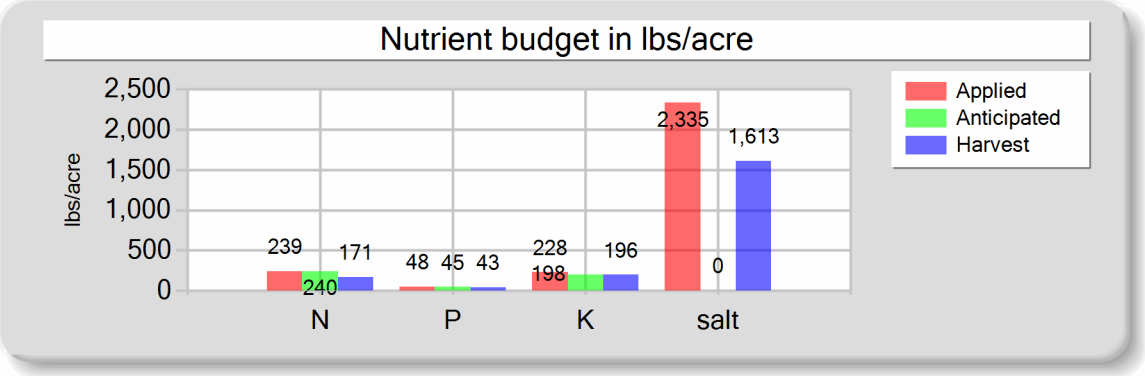
Fresh water applied	
47,483,630.00	*gallons*
1,748.66	*acre-inches*
16.04	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

MS-Almond - 06/10/2023: Corn, silage

Field name: MS-Almond Crop: Corn, silage Plant date: 06/10/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	224.02	48.41	228.12	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.17	0.00	0.00	2,334.89
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	239.19	48.41	228.12	2,334.89
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	171.14	42.79	195.83	1,612.69
Nutrient balance	68.05	5.63	32.30	722.21
Applied to removed ratio	1.40	1.13	1.16	1.45

Fresh water applied	
152,488,610.00	*gallons*
5,615.64	*acre-inches*
51.52	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

2022 Fall Corral Manure

Sample and source description: 2022 Fall Corral Manure

Sample date: 10/05/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 40.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,300.00	3,700.00	18,400.00							0.00
DL	0.01	0.01	0.01							1.00

Scraped Manure

Sample and source description: Scraped Manure

Sample date: 05/15/2023 Material type: Scraped material Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,800.00	10,500.00	43,900.00	21,200.00	15,400.00	14,000.00	6,800.00	27,000.00		42.10
DL	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01		1.00

Scraped Manure

Sample and source description: Scraped Manure

Sample date: 12/13/2023 Material type: Scraped material Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,300.00	5,900.00	27,800.00							0.00
DL	0.01	0.01	0.01							1.00

B. PROCESS WASTEWATER ANALYSES

2022 Lagoon Q4

Sample and source description: 2022 Lagoon Q4

Sample date: 11/22/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.80

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	142.00	110.00	0.00	0.90	26.90	327.00								5,770.00	2,820
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q1

Sample and source description: Lagoon Q1

Sample date: 03/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	113.00	94.00	0.00	0.70	42.60	321.00								3,640.00	2,050
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q2

Sample and source description: Lagoon Q2

Sample date: 05/11/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.90

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	180.00	137.00	0.00	0.50	54.40	334.00	154.00	113.00	397.00	2,110.00	0.00	13.40	408.00	5,590.00	1,940
DL	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	10.00	1

Lagoon Q3

Sample and source description: Lagoon Q3

Sample date: 07/25/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	147.00	133.00	0.00	2.30	41.40	304.00								4,600.00	2,190
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q4

Sample and source description: Lagoon Q4

Sample date: 12/13/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 8.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	153.00	128.00	0.00	1.50	24.40	200.00								4,900.00	2,520
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

C. FRESH WATER ANALYSES

CCID Canal

Canal

Sample description: Canal

Sample date: 10/17/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.70	0.00	0.70								328.00	200
DL	0.01	0.01	0.01								10.00	1

Domestic Well

DW

Sample description: DW

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	10.20	0.00	10.20								1,730.00	0
DL	0.01	0.01	0.01								10.00	1

Irrigation Well

Irrigation Well

IW

Sample description: IW

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	8.50	0.00	8.50								1,060.00	0
DL	0.01	0.01	0.01								10.00	1

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

MS1 - 10/08/2021: Alfalfa, hay

MS1 Alfalfa

Sample and source description: MS1 Alfalfa

Sample date: 07/21/2022 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 28.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	38,100.00	3,000.00	32,500.00		10.60
DL	0.01	0.01	0.01		1.00

MS2 - 10/08/2022: Oats, silage-soft dough

MS2 - 10/08/2022: Oats, silage-soft dough

Oats - MS #2

Sample and source description: Oats - MS #2

Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 51.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,400.00	2,300.00	19,300.00		9.60
DL	0.01	0.01	0.01		1.00

MS2 - 06/10/2023: Corn, silage

Corn MS #2

Sample and source description: Corn MS #2

Sample date: 09/22/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,100.00	2,400.00	12,600.00		6.10
DL	0.01	0.01	0.01		1.00

MS-Almond - 10/08/2022: Oats, silage-soft dough

Oats-MS-Almond

Sample and source description: Oats-MS-Almond

Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 57.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,000.00	2,300.00	12,000.00		8.20
DL	0.01	0.01	0.01		1.00

MS-Almond - 06/10/2023: Corn, silage

Almond Corn

Sample and source description: Almond Corn

Sample date: 11/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 56.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,400.00	2,600.00	11,900.00		9.80
DL	0.01	0.01	0.01		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

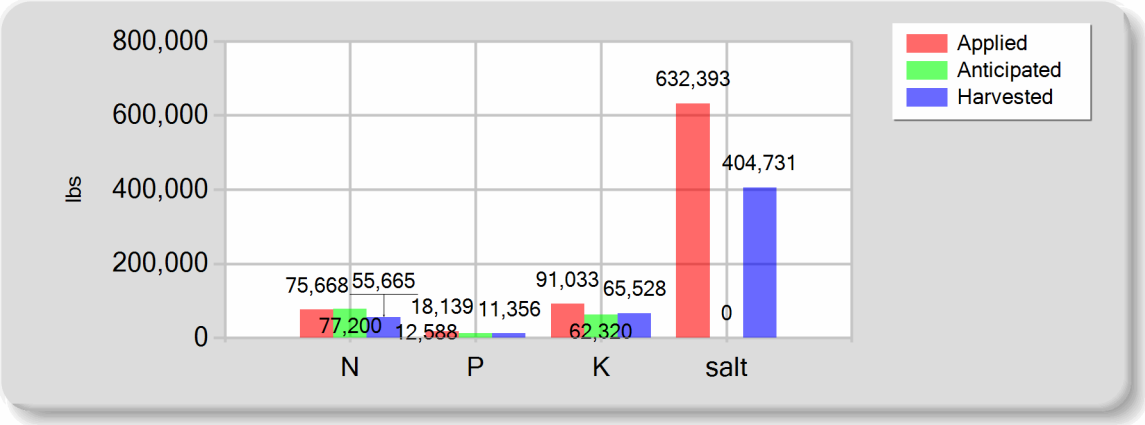
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

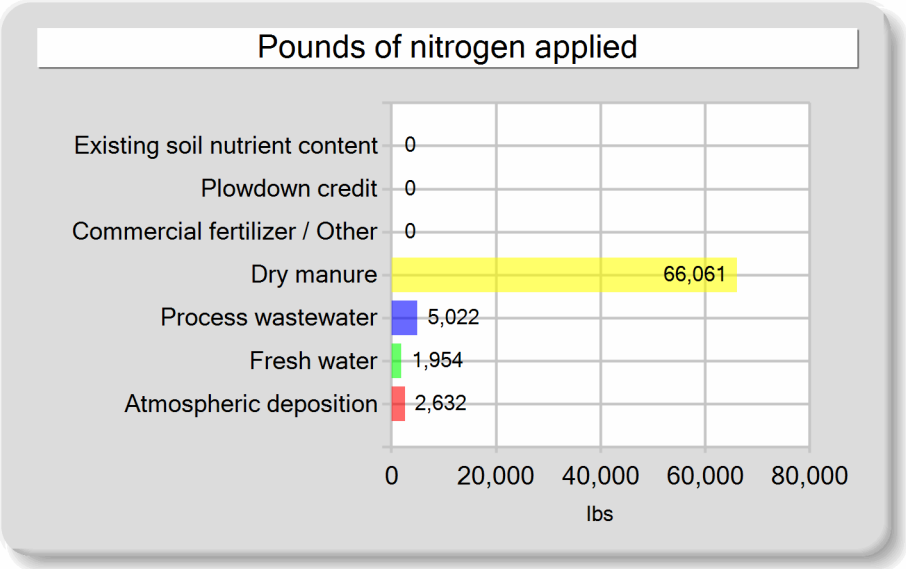
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

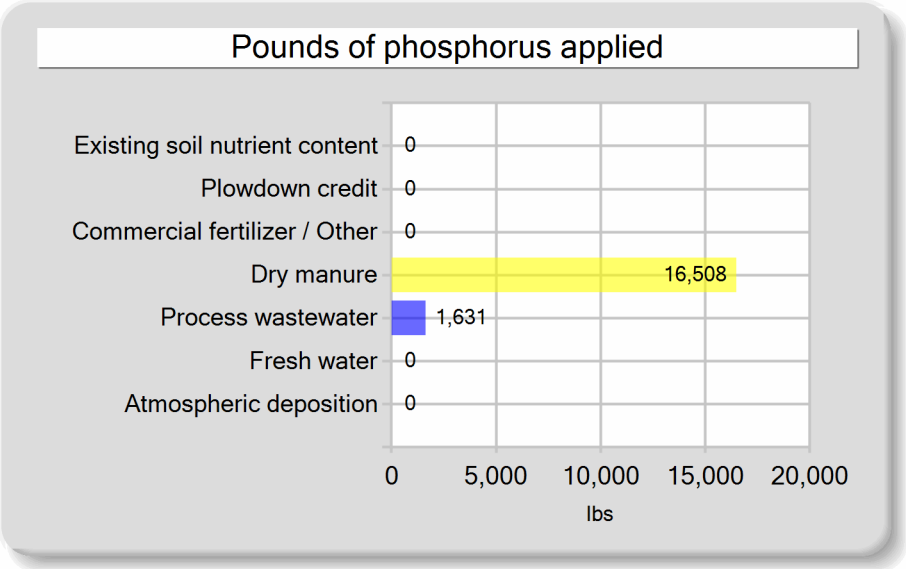
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	66,060.90	16,507.99	79,496.97	0.00
Process wastewater	5,021.82	1,631.28	11,536.13	74,173.70
Fresh water	1,953.77	0.00	0.00	558,219.42
Atmospheric deposition	2,632.00	0.00	0.00	0.00
Total nutrients applied	75,668.49	18,139.27	91,033.09	632,393.11
Anticipated crop nutrient removal	77,200.00	12,587.80	62,320.40	0.00
Actual crop nutrient removal	55,665.43	11,356.15	65,527.90	404,730.87
Nutrient balance	20,003.06	6,783.11	25,505.19	227,662.25
Applied to removed ratio	1.36	1.60	1.39	1.56

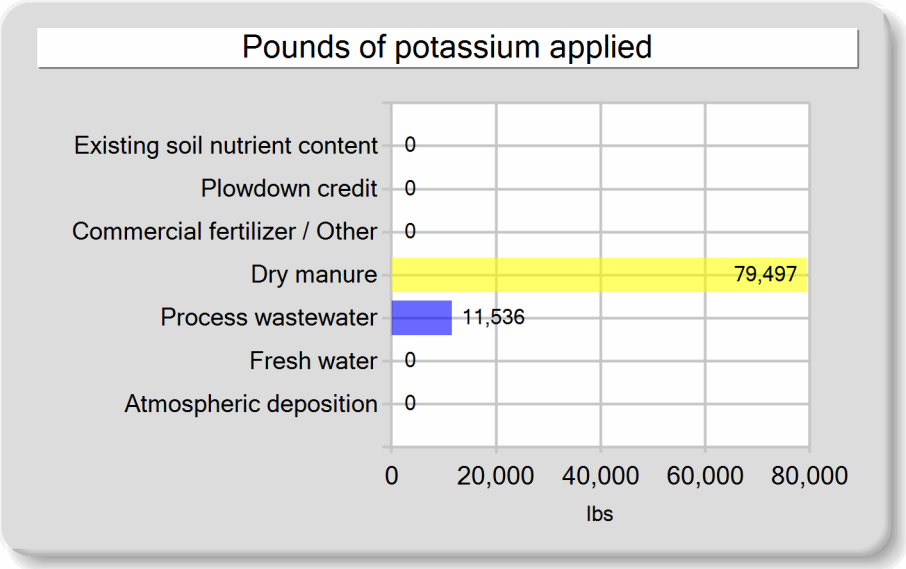
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

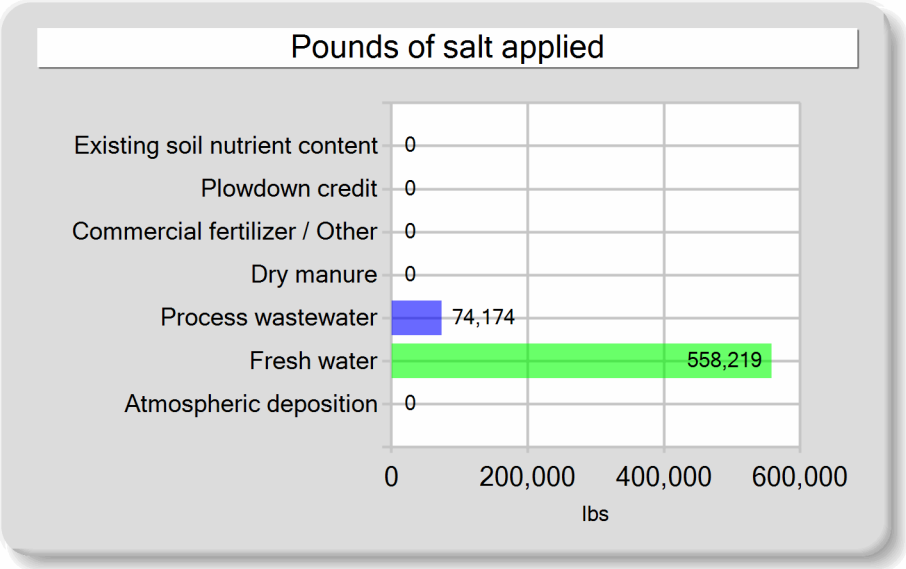


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

David Soares	David Soares
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Soares Dairy Farms Inc.	David Soares
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jul 1, 2024	Jul 1, 2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Report of Dairy Well Water Analysis

DELLAVALLE LABORATORY INC

Soares Mercy Springs Dairy
14515 Badger Flat Rd
Los Banos CA 93635
00-0022800 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L0988
Sampled By: M Dekeyrel
Requested By:
Submitted Date: 12/14/23
Reported Date: 12/21/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	pH at 25°C unit
1	DW	12/13/23	16:35	1730	1.73	10.2	ND		7.7
2	IW	12/13/23	16:35	1060	1.06	8.5	ND		7.3

***** = Field NH_4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Soares Mercy Springs Dairy
14515 Badger Flat Rd
Los Banos CA 93635
00-0022800 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L0988
Sampled By: M Dekeyrel
Requested By:
Submitted Date: 12/14/23
Reported Date: 12/21/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Nitrogen agricultural use Calcs calculated from nitrate-N lbs/AcFoot	lbs/AcInch
1	DW	12/13/23	16:35	**1730**	**1.73**	**10.2**	27.85	2.32
2	IW	12/13/23	16:35	**1060**	**1.06**	**8.5**	23.21	1.93

Water for Crop Use **Nutrient Management**	Total Salts	Total Salts	Nitrate Nitrogen
Low Levels	< 900	< 0.90	< 2.0
Significant Levels	900-2200	0.90-2.2	2.0-10.0
High Levels	2200+	2.2+	10.0+

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.
Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).
Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.
MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)
MDL = Method Detection Limit; **RL** = Reporting Limit, mg/L = ppm.
SM = Standard Methods for the Examination of Water and Wastewater
EPA = Environmental Protection Agency methods used unless otherwise indicated.

DELLAVALLE™
LABORATORY INC

Soares Mercy Springs Dairy
14515 Badger Flat Rd
Los Banos, CA 93635

Account# 00-0022800
Account Manager: Ben Nydam
Submitted By:

Received: 12/14/2023 14:20
Reported: 12/21/2023 10:20

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23L0988-01	DW	Ag Water	M Dekeyrel		12/13/2023 16:35
23L0988-02	IW	Ag Water	M Dekeyrel		12/13/2023 16:35

Default Cooler Temperature on Receipt °C: 5.2
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

Soares Mercy Springs Dairy
14515 Badger Flat Rd
Los Banos, CA 93635

Account# 00-0022800
Account Manager: Ben Nydam
Submitted By:

Received: 12/14/2023 14:20
Reported: 12/21/2023 10:20

Sample Results

Sample: DW
23L0988-01 (Water)

Sampled: 12/13/2023 16:35
Sampled By: M Dekeyrel

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.73**	mmhos/cm	0.01	1		12/18/23 17:24	SM 2510 B		BEL0701
Electrical Conductivity umhos	**1730**	umhos/cm	10.0	1		12/18/23 17:24	SM 2510 B		BEL0701
Ammonia (as N)	ND	mg/L	0.00	1		12/13/23 16:35	Field		BEL0630
Nitrate Nitrogen as NO3N	**10.2**	mg/L	0.1	1	10	12/17/23 02:14	EPA 300.0		BEL0665
Temperature	**25.0**	units	0.0	1		12/18/23 17:24	SM 4500-H+	H	BEL0701
pH	**7.7**	units	1.0	1		12/18/23 17:24	SM 4500-H+	H	BEL0701

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DELLAVALLE™
LABORATORY INC

Soares Mercy Springs Dairy
14515 Badger Flat Rd
Los Banos, CA 93635

Account# 00-0022800
Account Manager: Ben Nydam
Submitted By:

Received: 12/14/2023 14:20
Reported: 12/21/2023 10:20

Sample Results
(Continued)

Sample: IW
23L0988-02 (Water)

Sampled: 12/13/2023 16:35
Sampled By: M Dekeyrel

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.06**	mmhos/cm	0.01	1		12/18/23 17:25	SM 2510 B		BEL0701
Electrical Conductivity umhos	**1060**	umhos/cm	10.0	1		12/18/23 17:25	SM 2510 B		BEL0701
Ammonia (as N)	ND	mg/L	0.00	1		12/13/23 16:35	Field		BEL0630
Nitrate Nitrogen as NO3N	**8.5**	mg/L	0.1	1	10	12/17/23 02:34	EPA 300.0		BEL0665
Temperature	**25.0**	units	0.0	1		12/18/23 17:25	SM 4500-H+	H	BEL0701
pH	**7.3**	units	1.0	1		12/18/23 17:25	SM 4500-H+	H	BEL0701

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14515 Badger Flat Rd
Los Banos, CA 93635

Account# 00-0022800
Account Manager: Ben Nydam
Submitted By:

Received: 12/14/2023 14:20
Reported: 12/21/2023 10:20

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0665										
Blank (BEL0665-BLK1)					Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0665-BLK2)					Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0665-BLK3)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0665-BLK4)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0665-BLK5)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEL0665-BS1)					Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000		97.2	90-110		
LCS (BEL0665-BS2)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000		98.8	90-110		
LCS (BEL0665-BS3)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000		98.0	90-110		
LCS (BEL0665-BS4)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		99.3	90-110		
Duplicate (BEL0665-DUP1)		**Source: 23L0964-06**			Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	5.8		0.1	mg/L		5.8			0.639	10
Duplicate (BEL0665-DUP2)		**Source: 23L0989-02**			Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	1.3		0.1	mg/L		1.3			0.787	10
Duplicate (BEL0665-DUP3)		**Source: 23L1000-02**			Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	0.2		0.1	mg/L		0.2			0.612	10
Duplicate (BEL0665-DUP4)		**Source: 23L1005-02**			Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	3.7		0.1	mg/L		3.7			0.941	10
Matrix Spike (BEL0665-MS1)		**Source: 23L0964-06**			Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	10.8		0.1	mg/L	5.000	5.8	102	90-110		
Matrix Spike (BEL0665-MS2)		**Source: 23L0989-02**			Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	6.4		0.1	mg/L	5.000	1.3	102	90-110		

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14515 Badger Flat Rd
Los Banos, CA 93635

Account# 00-0022800
Account Manager: Ben Nydam
Submitted By:

Received: 12/14/2023 14:20
Reported: 12/21/2023 10:20

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0665 (Continued)										
Matrix Spike (BEL0665-MS3)			**Source: 23L1000-02**		Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	4.7		0.1	mg/L	5.000	0.2	91.6	90-110		
Matrix Spike (BEL0665-MS4)			**Source: 23L1005-02**		Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	8.7		0.1	mg/L	5.000	3.7	99.3	90-110		
Reference (BEL0665-SRM1)					Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	10.1			mg/L	10.00		101	90-110		
Reference (BEL0665-SRM2)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEL0665-SRM3)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEL0665-SRM4)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEL0665-SRM5)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		

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Los Banos, CA 93635

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Account Manager: Ben Nydam
Submitted By:

Received: 12/14/2023 14:20
Reported: 12/21/2023 10:20

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0701										
Blank (BEL0701-BLK1)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	5.5		1.0	units						
Blank (BEL0701-BLK2)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.7		1.0	units						
Blank (BEL0701-BLK3)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
pH	8.1		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Duplicate (BEL0701-DUP1)		**Source: 23L0985-02**			Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1.51		0.01	mmhos/cm		1.48			2.19	10
pH	7.7		1.0	units		7.7			0.130	10
Electrical Conductivity umhos	1510		10.0	umhos/cm		1480			2.19	10
Duplicate (BEL0701-DUP2)		**Source: 23L0989-03**			Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1.84		0.01	mmhos/cm		1.83			0.730	10
Electrical Conductivity umhos	1840		10.0	umhos/cm		1830			0.730	10
pH	8.7		1.0	units		8.7			0.00	10
Reference (BEL0701-SRM1)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	424			umhos/cm	426.0		99.6	90-110		
Reference (BEL0701-SRM2)					Prepared & Analyzed: 12/18/2023					
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEL0701-SRM3)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1010			umhos/cm	1000		101	90-110		
Electrical Conductivity umhos	1010			umhos/cm	1000		101	90-110		
Reference (BEL0701-SRM4)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1020			umhos/cm	1000		102	90-110		
Electrical Conductivity umhos	1020			umhos/cm	1000		102	90-110		
Reference (BEL0701-SRM5)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1020			umhos/cm	1000		102	90-110		

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LABORATORY INC

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14515 Badger Flat Rd
Los Banos, CA 93635

Account# 00-0022800
Account Manager: Ben Nydam
Submitted By:

Received: 12/14/2023 14:20
Reported: 12/21/2023 10:20

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0701 (Continued)										
Reference (BEL0701-SRM5)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity umhos	1020			umhos/cm	1000		102	90-110		
Reference (BEL0701-SRM6)					Prepared & Analyzed: 12/18/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0701-SRM7)					Prepared & Analyzed: 12/18/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0701-SRM8)					Prepared & Analyzed: 12/18/2023					
pH	4.0			units	4.000		101	97.5-102.5		

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12/14/23 14:20 23L0988

JG

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 ~~800 228-9896~~ • Fax 559 268-8174

WATER WORK REQUEST

Acct No. / Cons.

Bill To: 22800 | 8

Purchase Order No. _____ Results Needed By _____

Client: Soares - Mercey Springs Dairy
Address: 14515 Badger Flat Road
City, State, Zip: Los Banos, CA 93635
Phone _____ Fax _____
Cell/Email _____

Copy to: Cardoso Ag Sevices - cas.labs@yahoo.com

Requested by _____

Ranch _____

Date sampled: 12/13/23

Sampled by: M DeKeyrel

[X] QA/QC Document [X] Copy of Chain [ ] RWQCB

No. of Samples _____ No. Bottles 2

Water Type: [ ] Drinking [ ] Wastewater
[ ] Ag Water [ ] Ground Water [ ] Mon. Well
[ ] Supply Water [ ] Other _____

Analysis and Bottles Required: *(Please Indicate Analysis)*

(✓) DWW1: (EC, pH, NO_3-N, **NH_4-N Field Test)**
(1) l L plastic, unpreserved (white)

( ) DWW2: (DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS)
(1) l L plastic, unpreserved (white)

( ) DCW1: (EC, NO_3-N, TDS)
(1) l L plastic, unpreserved (white)

( ) DPW1: (EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK)
(1) l L plastic, unpreserved (white)

( ) DPW2: (DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl)
(1) l L plastic, unpreserved (white)

( ) Other _____

DESCRIPTION OF SAMPLES		Date Sampled	Time Sampled	Field NH4-N (mg/L)	Received Temp °C
1. DW	Sampled From:	12/13	4:35	Ø	5.2
2. IW	Sampled From:	12/13	4:35	Ø	4.5
3.	Sampled From:				
4.	Sampled From:				
5.	Sampled From:				
6.	Sampled From:				
7.	Sampled From:				
8.	Sampled From:				
9.	Sampled From:				
10.	Sampled From:				

IR Thermometer SN: 221314357
Correction Factor: 0°C
Calibration Due: 03/06/2024
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	M DeKeyrel	CAS INC	12/13/23 @ 4:36	12/14/23 12p
Second	CSM	DLI	12/14/23 12pm	12/14/23 2:15pm
Third				
Fourth	[illegible]	DLI	12/14 14:20	

I guarantee that as the client, or on behalf of the client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24 %) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (cal). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through cal under its Rules and Procedures. The prties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Sampling Hrs ____ Miles ____ Consulting ____

Amt Paid ____ Rec By ____ Check No. ____

Shipping
$ _____ In
$ _____ Out
Date _____

Signature _____

Sample received in cooler with ice?
[ ] Yes [ ] No

ctt:update 2020

Shipping Information: Shipped In ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐ Other ☐ _____

Container: Ice Chest ☒ Box ☐ None ☐ **Refrigerant:** Wet Ice ☒ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

12/14/23 14:20 23L0988

Type of Container(s) Received		Sample Number 1	2	3	4	5	6	7	8	9	10
Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic										
	* pH Value										
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
"	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ + MCAA										
	40 mL AG VOA, $Na_2S_2O_3$										
	40mL AG VOA unpreserved (White) (Set of 3										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

SEND OUT